Wellbore DDMS

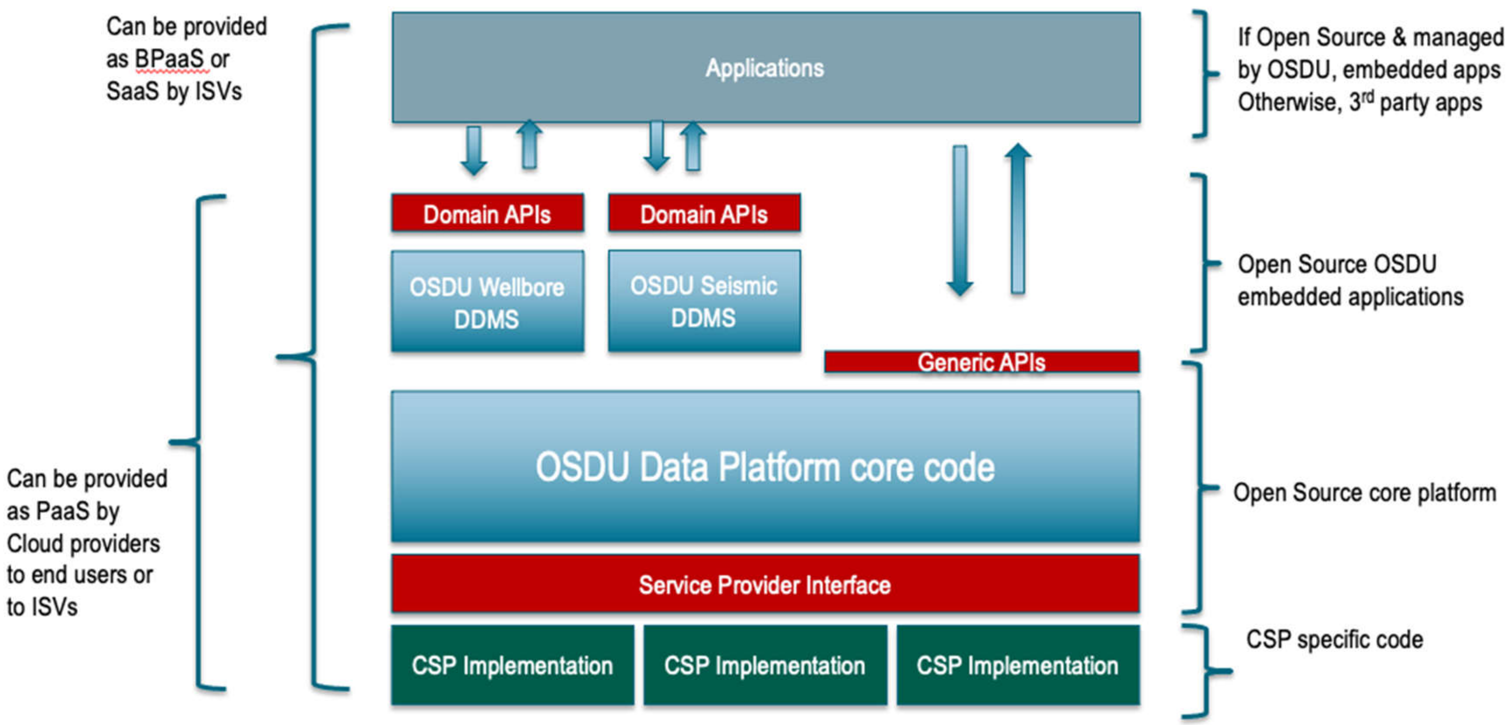
» Introduction
» Product description
» Architecture
» Initial scope

Expanding Data Platform to multiple domains

- Ability to develop domain extensions in parallel
  - Split domain data flows, domain storage/access concerns
- Support domain developers thru type-safe interfaces instead of generic core services
- Address domain specific concerns
  - Core Platform governs minimum principles across domains and shared services

Velocity and Autonomy is key

OSDU Data Platform Software Components



What is a Domain Data Management Service aka DDMS?

A service that persists data of a specific domain and provide access through optimized domain APIs.

It is governed by the platform but is developed and evolved independently.

A DDMS :

- Leverage the core services
- Adhere to the usage patterns and common behavior
- Provide highly optimized storage & access for bulk data, with highly opinionated API's delivering the data required to enable ***domain*** workflows

Example for the WellLog entity

Before

ID	987654
Kind	OSDU:WPC: WellLog
Wellbore ID	123456
Projected CRS ID	EPSG : 4326
Null Value	-999.25
…….	…….

Measurement	MD	GR	DENS	NEUT	…….
Unit	FT	GAPI	G/CC3	v/v	…….

With Wellbore DDMS

ID	987654
Kind	OSDU:WPC: WellLog
Wellbore ID	123456
Projected CRS ID	EPSG : 4326
Null Value	-999.25
…….	…….

Measurement	MD	GR	DENS	NEUT	…….
Unit	FT	GAPI	G/CC3	v/v	…….
	100	30	2.21	0.15	……
	100.5	32	2.23	0.17	……
	101	35	2.26	0.18	……
	101.5	37	2.25	0.2	……
	102	36	2.27	0.22	……

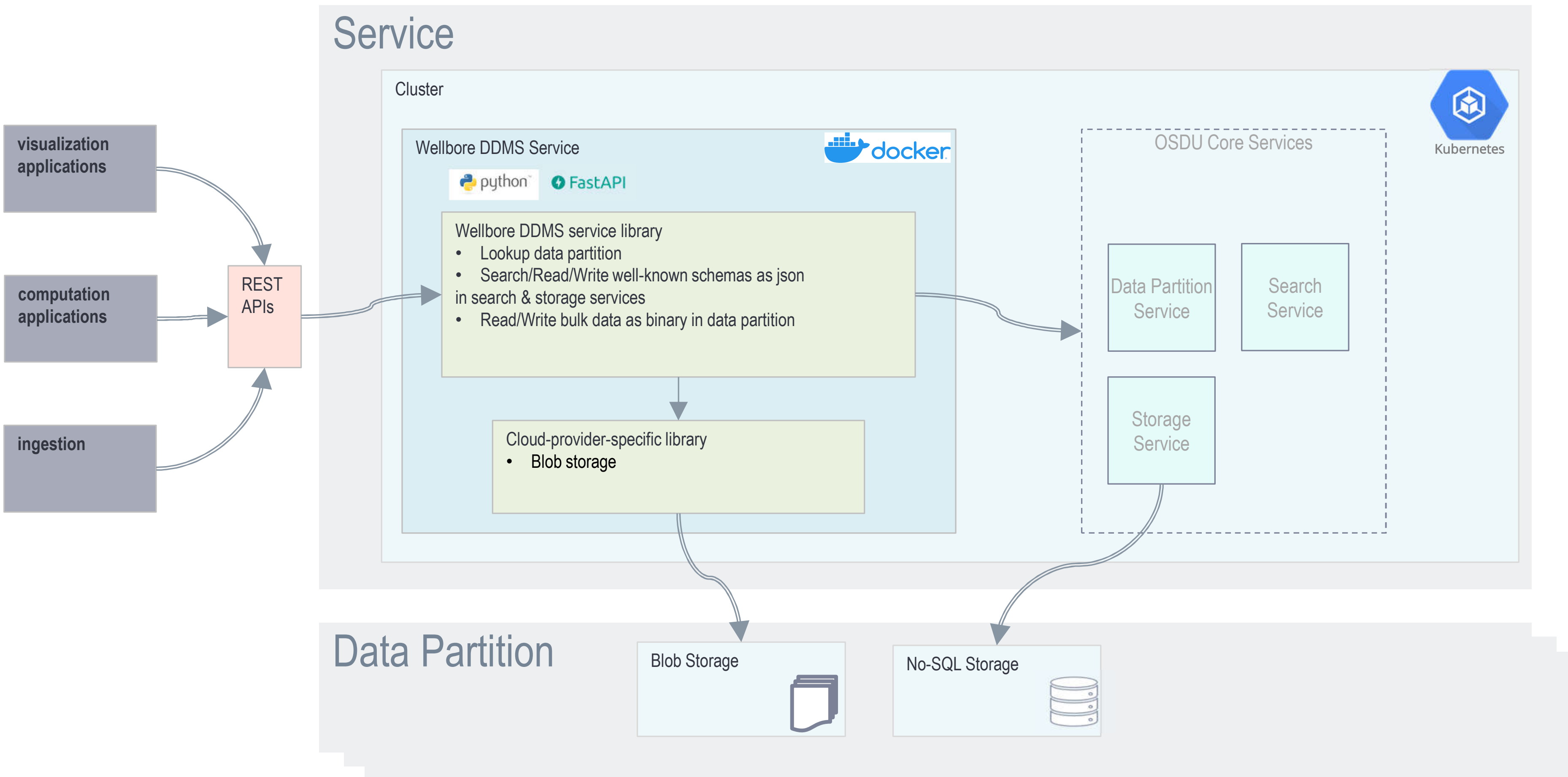
Bulk storage + Relevant end points

- Give me **WellLog** entities associated with **Wellbore** 123456
- Give me **ONLY** WellLog that contains GR & DENS & RES & …
- Give me **bulk data** of only GR & DENS & …
- Give me **bulk data of GR & DENS ONLY between** 0-1000 FT

Schlumberger

Schlumberger-Private

Architecture



Schlumberger

Scope of contribution for the consortium

Data Access

OSDU R3 model:
Master data:
- Well & Wellbore

Work Product Component:
- WellLog
- WellboreTrajectory
- WellboreMarkerSet

Convenient Queries

I am looking for :
All Wellbores
All Curves

All Curves from WellLog X
All Curves from Wellbore Y

Wellbores in geographic area
WellLogs from Wellbores with specific attributes
Curves from Wellbores with specific attributes
Curves from WellLog with specific attributes
Curves with specific attributes

Data services

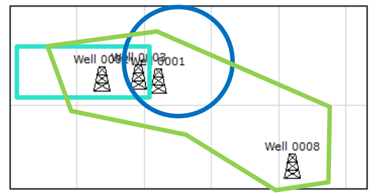
Log statistics
- Statistical information per Curve \ Column

Log recognition
- Assign a Curve to a family based on a catalog of rules
- Replace default catalog by a user catalog

More details in next slides

Convenient filters and queries (pending feasibility & data model upgrade)

Common access patterns

Wellbores in a specific **geographical area**	Based on rectangle, circle or polygone	
WellLog from **Wellbores** with specific attributes	**WellLogs** from Horizontal **Wellbores**	If **WellboreTrajectoryType==Horizontal** give me all the **WellLogs**
Curves from **Wellbores** with specific attributes	**Curves** from Horizontal **Wellbores**	If **WellboreTrajectoryType==Horizontal** give me **Gamma Ray & Density Curves**
Curves from **WellLogs** with specific attributes	**Curves** from **WellLogs** between 1000-2000 ft	If **TopMeasuredDepth>1000 FT & BottomMeasuredDepth<2000 FT**, give me **Gamma Ray curves**
Curves based on their **attributes**	**Curves** assigned to **Porosity**	Give me **Curves** where the **LogCurveFamilyID==Porosity**

Log Recognition

Business question : I want to calculate XYZ, where do I relevant data ?

Well 1	
APL	ft3/ft3
ECGR	GAPI
DEN	g/cm3
RES_D	ohm.m

Well 2	
APS	v/v
GR	gAPI
DENS	g/cc
ILD	ohm.m

Well 3	
CNC	V/V
CGR	gAPI
RHOB	g/cm3
RW	ohm.m

Well 4	
FTEMP	degC
RW	V/V
SWE	%
SWT	%

Well 5	
CALI	IN
GR_NORM	gAPI
RHO8	G/C3
AT90	ohm.m

Measurement
Porosity
Gamma Ray
Density
Resistivity

Well 1	
APL	ft3/ft3
ECGR	GAPI
DEN	g/cm3
RES_D	ohm.m

Well 2	
APS	v/v
GR	gAPI
DENS	g/cc
ILD	ohm.m

Well 3	
CNC	V/V
CGR	gAPI
RHOB	g/cm3
RW	ohm.m

Well 4	
FTEMP	degC
RW	V/V
SWE	%
SWT	%

Well 5	
CALI	IN
GR_NORM	gAPI
RHO8	G/C3
AT90	ohm.m

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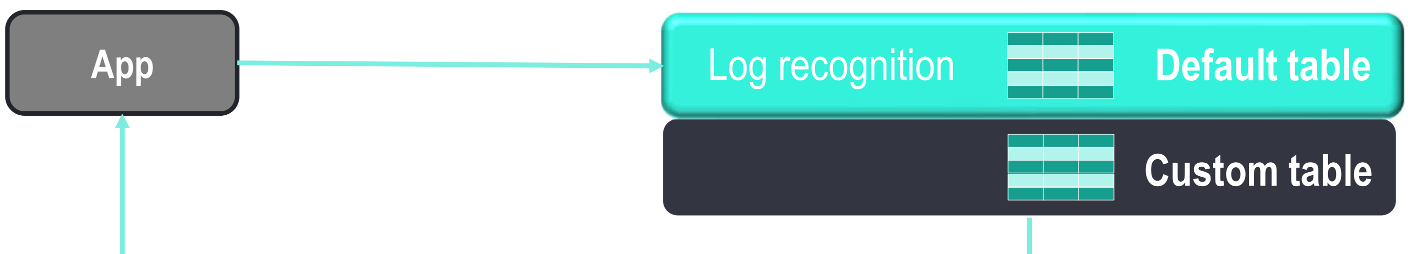
Log Recognition

Name, Unit

GR_FINAL, gAPI

RHOG, g/cc3



Main Family \ Family

Gamma ray \ Gamma ray

Density \ Grain Density

Requirements to ensure data is managed consistently across the platfrom

All raw data is preserved (via ingestion framework)

Ingestion and retention minimize data loss.

Data is globally identifiable (Via Storage ID's)

Context specific data identity prevents leveraging data.

Data is immutable

Metadata without exception. By exception, transient data is subject to cost considerations. Versioning preferred.

Data is access controlled (via Entitlements)

Only authorized identities have access to data

Data is governed for right of use (Compliance)

Data compliance is enforced in all entry/exit points.

Data is discoverable (Via Search & Index)

All data is indexed

Data is consumable (via the domain API's)

Data domain management service needs to provide a standardized way of consuming its data

Improved data is new data

Enrichment is a workflow and results in new data.

Data lineage is tracked

All transformations and workflows must provide lineage. Apps needs to fill the ancestry.parents[] and\or data.LineageAssertions[]

Data is referenced

Copies of data are possible in different consumption zones (e.g., data ponds) but domain data management stores are the ultimate sources of truth.

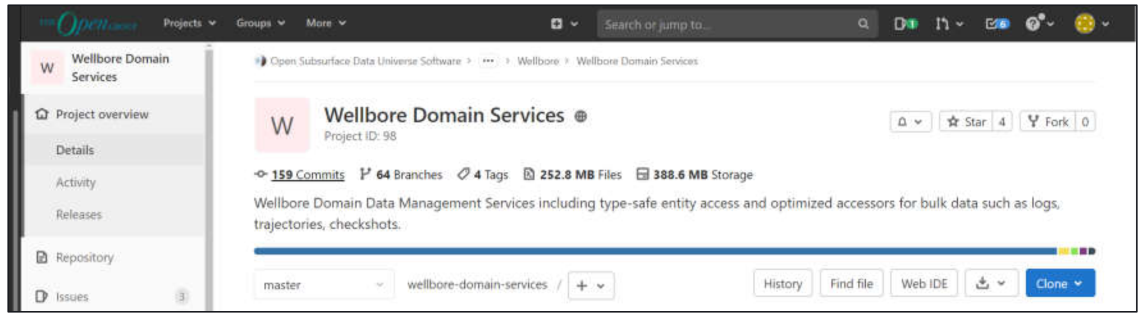
API Demo

https://saas.managed-osdu.cloud.slb-ds.com/api/os-wellbore-ddms/docs#/

Community

Open-source

Repository



https://community.opengroup.org/osdu/platform/domain-data-mgmt-services/wellbore/wellbore-domain-services

Slack

#10_3_r3_ddms